Recruiting

Project 08 Midterm: Final Motorized Sound Machine

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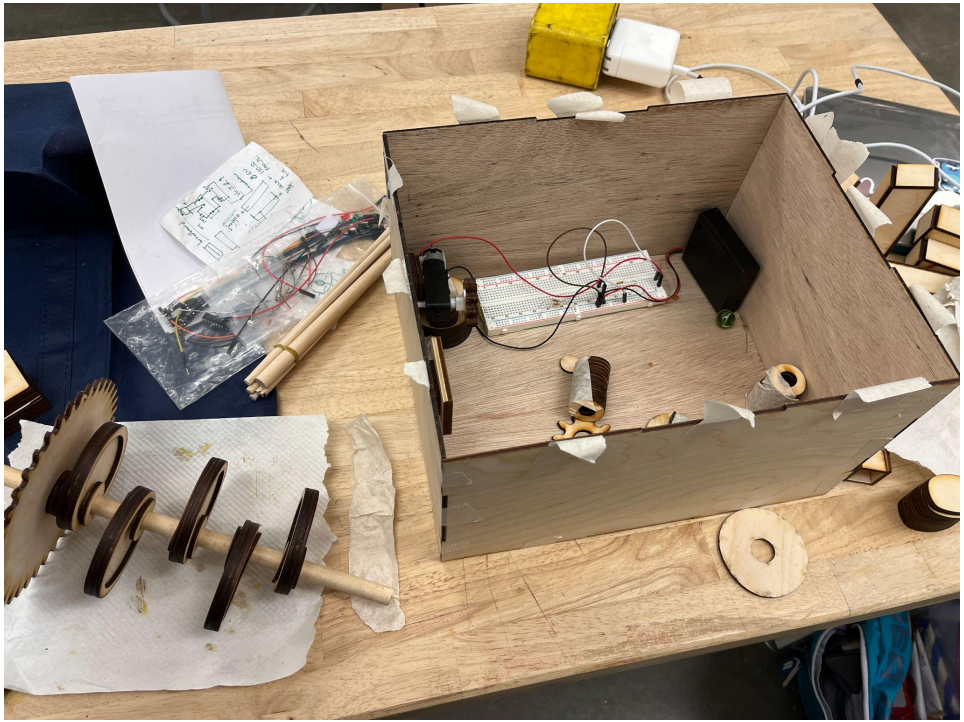
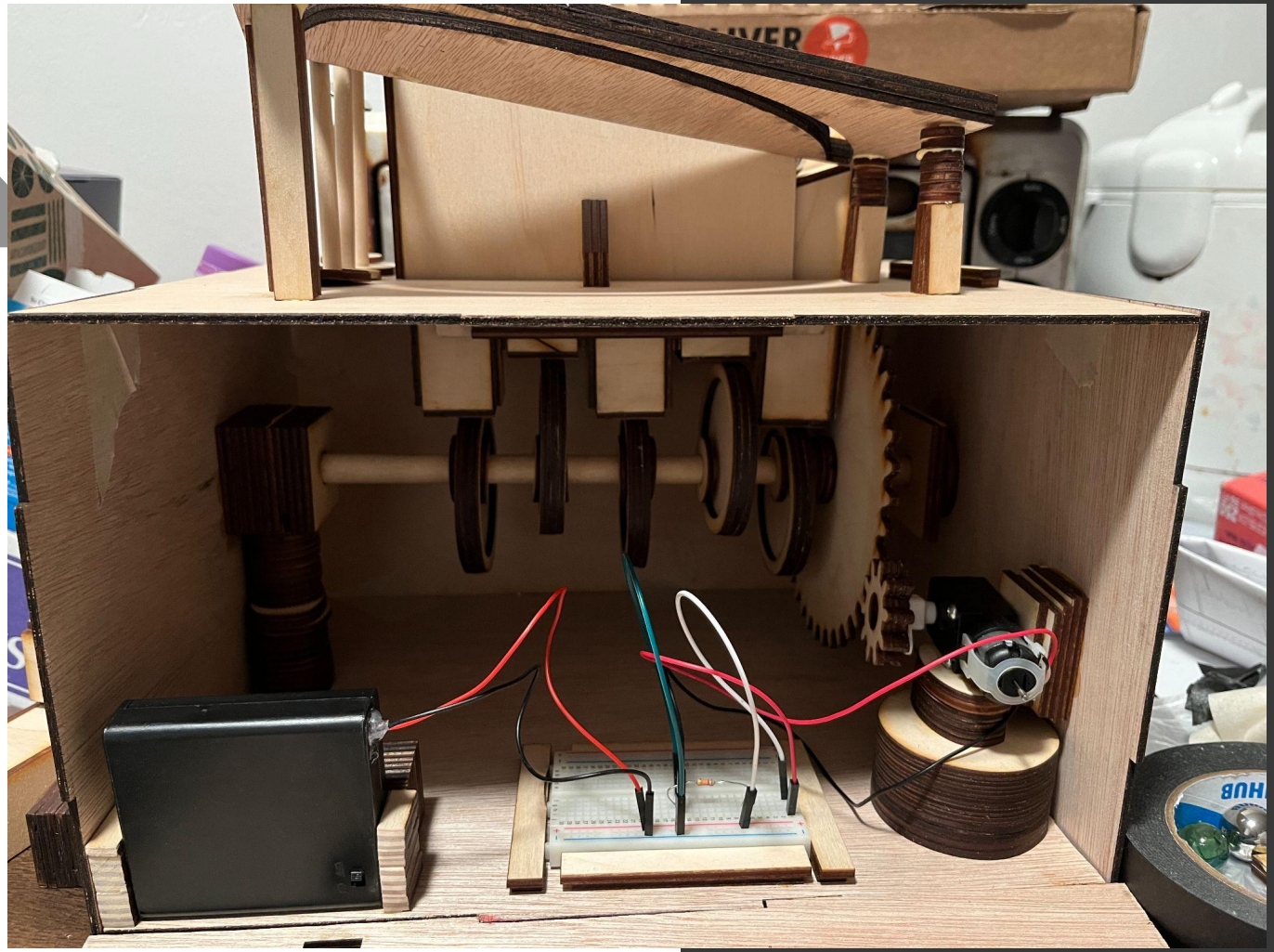
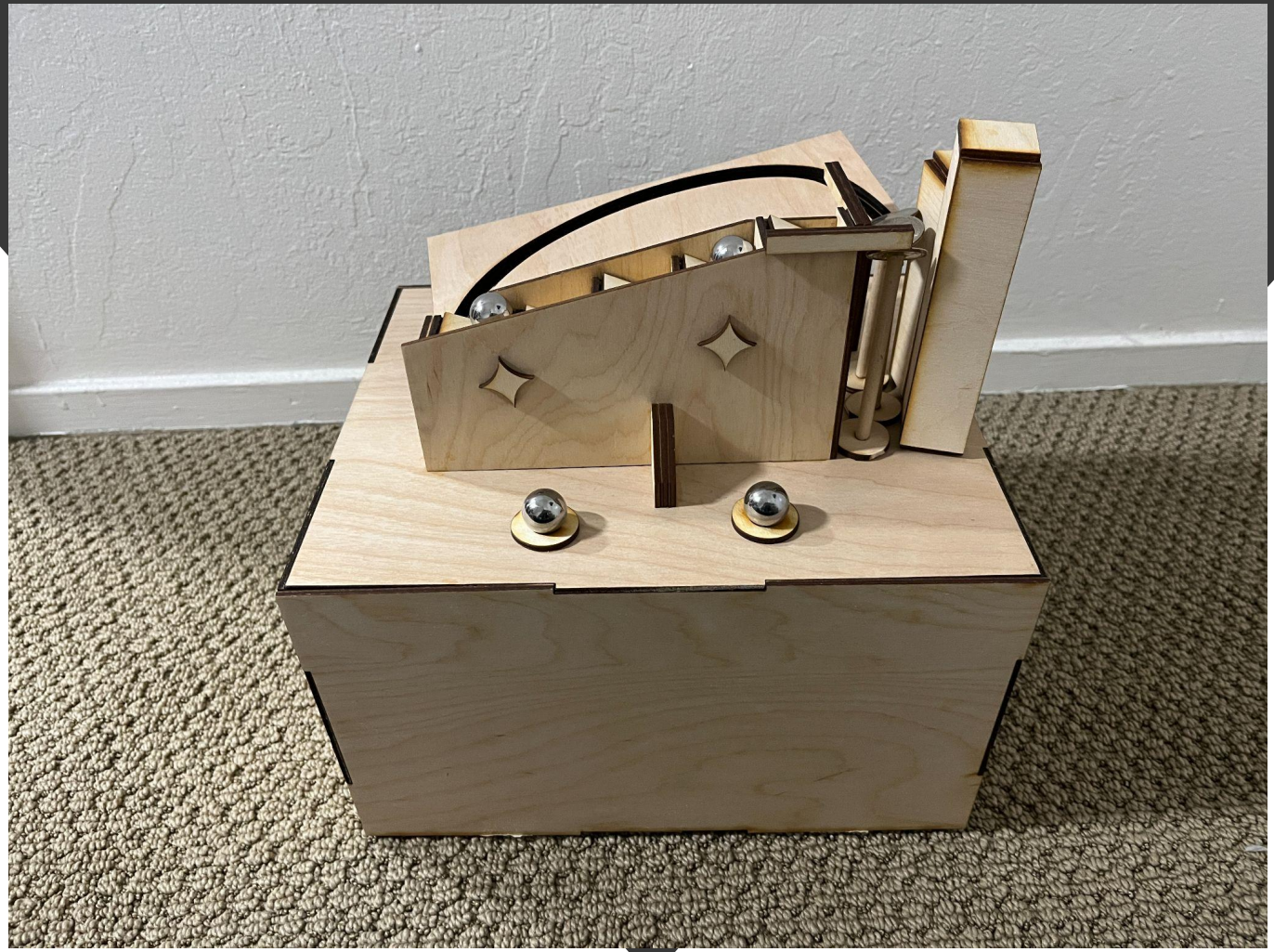
DESCRIPTION

An automatic staircase to make ball bearings ascend then fall onto quarters to create a musical rhythm, then repeat the process.

Video Link:
https://www.youtube.com/channel/UC2yV-KrUSXTSKdI8N-k9P4Q







PROCESS

The automatic staircase functionality was still the most difficult part of the project since there was no way to test whether the mechanism worked until gluing down all of the parts. The design of each staircase kept changing since there were additional factors to consider, such as the weight of the staircase since the dowel and cams would be supporting all of the stairs.

The speed of the staircase ascending was too fast so there needed to be a resistor to slow down the speed, but that reduces the power of the motor which is bad since the stairs are heavy. To combat this, the power of the motor is critical so the staircase just moves quickly.

I was unable to incorporate the original design of the model which included the ball bearings hitting mechanical keyboard switches to create a musical rhythm. There wasn't enough time and not enough room on the surface to incorporate the different position of the switches. Switching the keyboard switches to coins was a good replacement since there was still a small sound that came from the clinking of the two metals.

Bill of Materials

- 1 - ½” Wooden dowel
- 1 - Right Angle Gear Motor
- Battery box
- Breadboard
- 1- 20 ohm Resistor
- 3 - Quarters
- Birch plywood
- Skinny dowels
- Acrylic plastic
- Wood glue
- Handsaw
- Sandpaper
- Paste wax
- Laser cutter
- X-acto knife
- Black Gaffers Tape
- Ruler

Motorized mechanism

My motorized mechanism involves using a right angle motor attached to a dowel with cams in opposite directions, allowing the stairs to move marbles up each step in a fluid motion. The motor and battery box provided enough power to support the stairs and cams, causing the stairs to ascend at a fast rate.

However, I want the stairs to move a tiny bit slower to draw in the audience's attention so I will be adjusting the speed of the stairs in the future. There is the addition of a small and large gear to make the dowel turn slower.

REFLECTION

SURPRISES ENCOUNTERED

I thought that creating the staircase would be fairly straightforward because it only involves using cams and a dowel. However, I was scared to glue everything into place because there was limited plywood and tape wasn't strong enough to test whether my designs had worked. The power of the motor was also surprising because there was too much power, causing the steps to turn too fast.

FRUSTRATING PARTS

Getting the gears to turn smoothly was very frustrating because I couldn't have the dowel support too much weight, otherwise it wouldn't turn. On the other hand, I needed the large gear to reduce the speed of the staircase. I attached a small gear to the motor but since it was misaligned, it was hard positioning the two gears to contact smoothly and turn together. Positioning all of the stairs so they wouldn't fall through the hole was also difficult.

DIFFERENT APPROACHES

I would try to find a different circuit design so I wouldn't need to create the large gear. While the addition of the gears allowed the dowel to turn slower than its original speed, it made the box a lot bigger than it needed to be, so there is a lot of empty space. I would also make the length of the box longer to allower for the spacing of the coins to be more evenly distributed. I would also design the back of the box to have hinges so the back can be removable.

THANK YOU!